Project 1: Calibration

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Abstract

The project consists in solving the following problems regarding camera calibration:

1. Find intrinsinc parameters K and for each image, the extrinsic parameters R,t.
2. Computing the total reprojection error for all the grid points for an image selected from the calibration dataset.
3. Superimpose an object to the calibration image in 25 images and check the correctness by visual inspection.
4. Minimize the reprojection error using Maximun Likelihood Estimation.
5. Analize the standard deviation of the principal point as a function of the number of images used for calibration
6. Add radial distorsion compensation to the basic Zhang's calibration procedure
7. Compute the total reprojection error with radial distorsion compensation and make a comparison to the one without compensation

1 Description of the approach

The procedure will be described for each point above :

1. Zhang method is based on homography[1], so for each image we need to calculate the related homography, finding the mapping from the checkerboard plane to the image plane. I have built a list containing the corners and filling it for the particular image with the function cv2.findChessboardCorners(). Another list is created to contain all the homographies. To find them, iterate over each corresponding corners and by constructing the matrix below on the left and solving with SVD, the solution is the right singular vector corresponding to the smallest singular value[2].

$$\begin{bmatrix} m^\top & 0^\top & -um^\top \\ 0^\top & m^\top & -vm^\top \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = 0.$$

Then the matrix is normalized, dividing each element for $H[3,3]$ and the homography is appended to the list containing the homographies. Now, in order to find the calibration matrix K, we need to construct and solve the system $Vb = 0$ where $b = [B_{11}\ B_{12}\ B_{22}\ B_{13}\ B_{23}\ B_{33}\ ]^T$ and $B = (KK^T)^{-1}$. Again the solution is the right singular vector corresponding to the smallest singular value , so b and then building up B. Exploiting the Cholesky factorization of B, we find finally K.
Lastly for each image the extrinsic parameters are calculated. Two lists, one for R and one for t are built considering for every image the corresponding homography taken from list containing all of them and considering the following equations.

$$r_1 = \lambda K^{-1} h_1 \ r_2 = \lambda K^{-1} h_2 \ t = \lambda K^{-1} h_3 \ r_3 = r_1 \times r_2$$

Then the closest in frobenius norm is taken to have it orthogonal and is appended to the list.

2. Selecting a particular image, in this case the one identified by the ID16, the points are projected through the matrix

$$P = K[R\,t]$$

so the relative rotation and translation matrix are chosen from the lists built before. Then each point relative to the checkerboard frame is projected obtaining the image in Figure 1 and the calculation of the reprojection error is performed considering the points detected by cv2.findChessboardCorners.

$$\epsilon(P) = \sum_{i=1}^{m} (\frac{p_1^T m_{(i)}}{p_3^T m_{(i)}} - u_{(i)})^2 + (\frac{p_2^T m_{(i)}}{p_3^T m_{(i)}} - v_{(i)})^2$$

3. The object chosen to be superimposed in the first 25 images is a cylinder with a radius of 33 mm and a height of 55 mm. The cylinder geometry is constructed by defining points for the top and bottom(discretized into segments to approximate the curve), its center and the vertical lateral struts. To project these sets of points, every 3D point (X,Y,Z) is multiplied by the image-specific projection matrix P to obtain homogeneous coordinates (u,v,w). These are then normalized by dividing by w to yield the final pixel coordinates. The resulting projections are shown in Figure 2.

4. Using the maximum likelihood estimation suggested in [1], the matrix K and the image i related matrix R_i and t_i can be further refined by minimizing the following:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} ||m_{ij} - \hat{m}(K, R_i, t_i, M_j)||^2 \tag{1}$$

For each image 88 points have been chosen as before, and a total number of 25 images, so $m = 88$ and $n = 25$.

As first analysis,the simpler model will be considered, optimize assuming zero skew ($\theta = 90$).

$$K = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

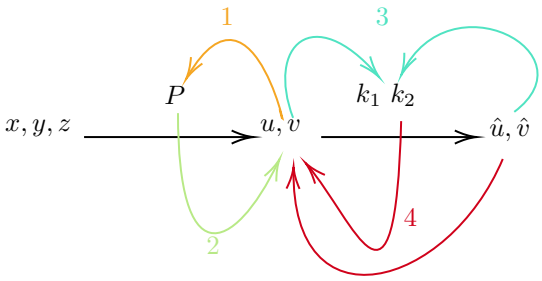
The vector of parameters that will be optimized according to the above functional by calling the following function $functionleast_squares(bundle_adjustment_residuals, x0 = x0, method =' lm', x_scale =' jac', args = (all_obj_points, all_img_points), verbose = 1)$ is constructing initially by stacking elements belonging to the matrices K, R_i, t_i previously calculated. In particular it will take the form $[\alpha_u, \alpha_v, u_0, v_0, rx_0, ry_0, rz_0, tx_0, ty_0, tz_0, rx_1, ...]$. The first 4 elements belong to the calibration matrix, shared by all the images and the others, rotational and translational values. Regarding rotations, the matrix R is expressed by Rodrigues formula by a vector of three elements. So the optimization problem consists of finding the vector of dimension $4+25\times6 = 154$ elements. As can be seen from the results, the projected points with these new optimized matrices are closer to the real corner of the chessboard, as shown in figure 3, indeed the reprojection error and consequently RMSE is less than before. Let's analyze what happens if we consider the general model, where matrix K has the following form:

$$K = \begin{bmatrix} \alpha_u & \alpha_u cot\theta & u_0 \\ 0 & \alpha_v / sin\theta & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Another parameter will be added to the vector to be optimized so for a total of 155 elements. The obtained matrices are then apply again to the image ID16. The results are show in table 1, for the selected image, the reprojection error and RMSE ate better when using the general model, as we could expect. The figure regarding the projection with the general model is not shown because it is very similar to the previous one and the details are very small to be compared visually.

5. The experiment to analyze how the number of images used influences the calibration is performed by considering a subset of 3 images to 25, randomly picked and for each subset a total of 50 trials. The results are in shown in figure 4.

6. In order to add the radial distorsion compensation we must estimate the radial distorsion parameters k_1, k_2 in iterative fashion as shown in the below diagram.



The iteration works in this way, observed (distorted) image coordinates are obtained using cv2.findChessboardCorners(). Subsequently, the projection matrix P (along with intrinsics K and extrinsics (R, t) is estimated via Zhang's method. The 3D object points are then projected onto the image plane using this linear model to obtain ideal, undistorted coordinates (u, v).

The radial distortion parameters k_1 and k_2 are estimated by solving a linear least squares problem that fits the radial model to the discrepancy between these ideal linear projections and the original observed distorted points. The model is described in:

$$\begin{cases} \hat{u} = (u - u_0)\left(1 + k_1 r_d^2 + k_2 r_d^4\right) + u_0, \\ \hat{v} = (v - v_0)\left(1 + k_1 r_d^2 + k_2 r_d^4\right) + v_0. \end{cases} \tag{2}$$

Finally, the original observed points are numerically undistorted using the estimated k_1,k_2 (via Newton method) to generate "straightened" coordinates; these refined coordinates serve as the input in the subsequent iteration. For the experiments I used 80 images and number 800 iterations. Using the same number of images as in the previous MLE experiment(25 images) was not enough, at the convergence the obtain reprojection erros was to high.
At iteration 800 the obtained value are : $k_1 : 0.05607$ $k_2 : -0.17637$. Results are in figures 5, 6, 8.

7. As already stated the reprojection error was too high when using only 25 images. Then I have decided to use all the images available , 80 images. Obviously the time needed to run the algorithm was far more longer than the previous ones but the results obtained are better, with a reprojection error of 7.62 (obviously with compensation of the radial distorsion parameters obtained in the previous estimation). Its behaviour is described in figure 7.

2 Results

Table 1: Comparison of the value $\epsilon(P)$ and RMSE

	No MLE	MLE simple model	MLE general model	Radial distorsion compensation
$\epsilon(P)\,[px^2]$	42.62	10.33	9.35	7.62
$RMSE\,[px]$	0.69	0.34	0.32	0.29

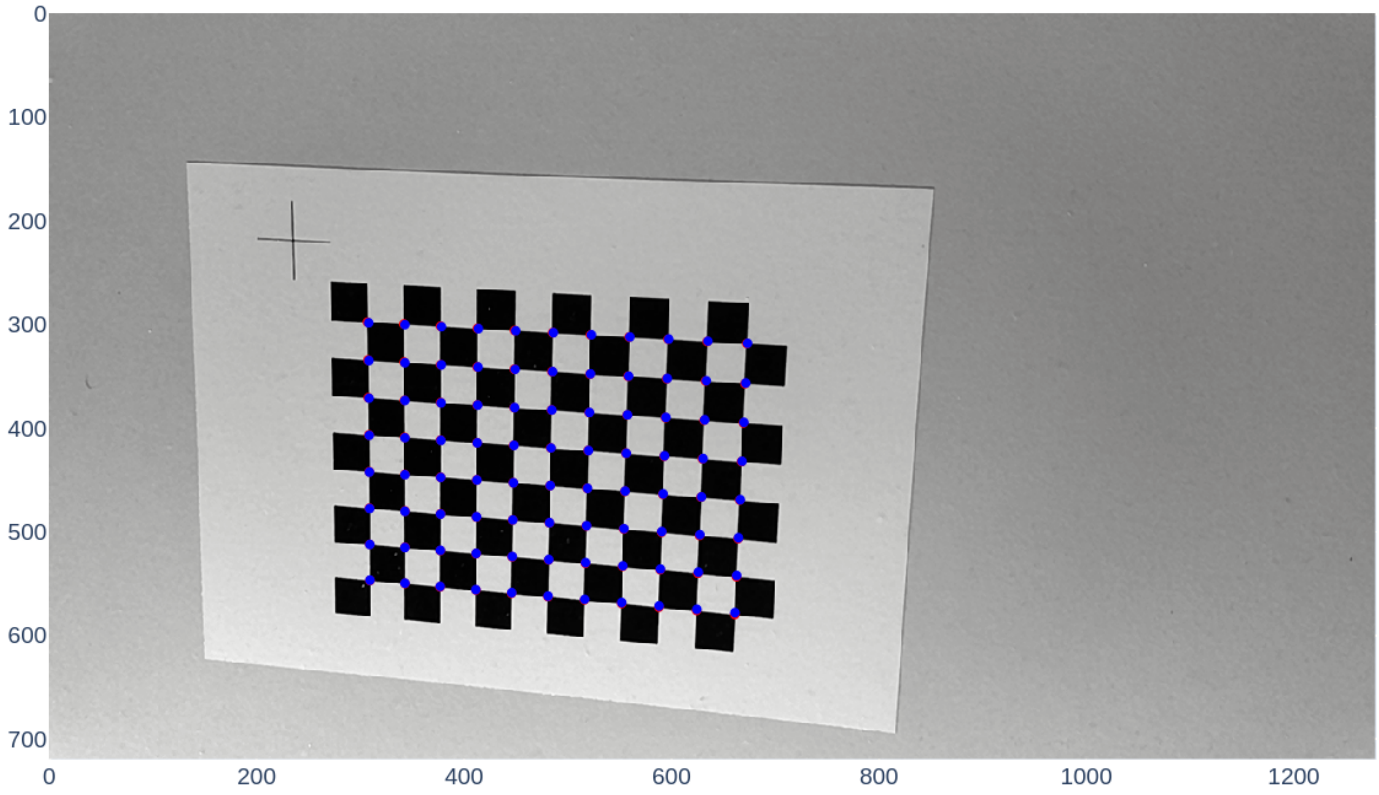


Figure 1: Corners (red) and projected points (blue)

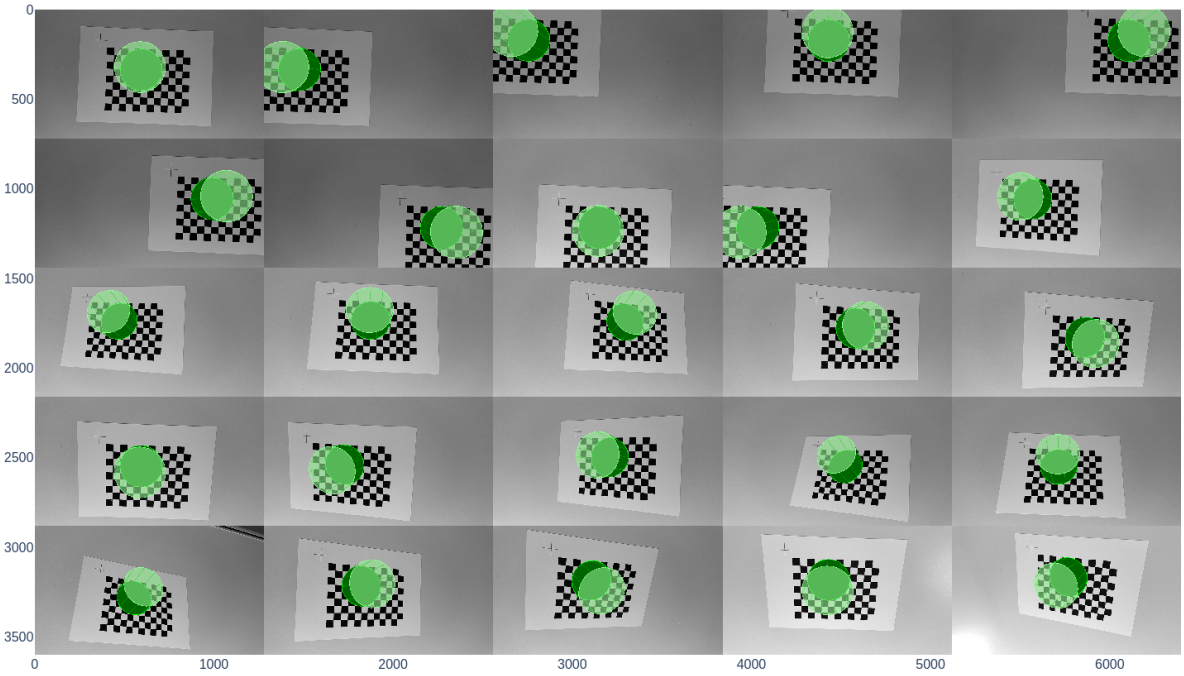


Figure 2: Projected Cylinders

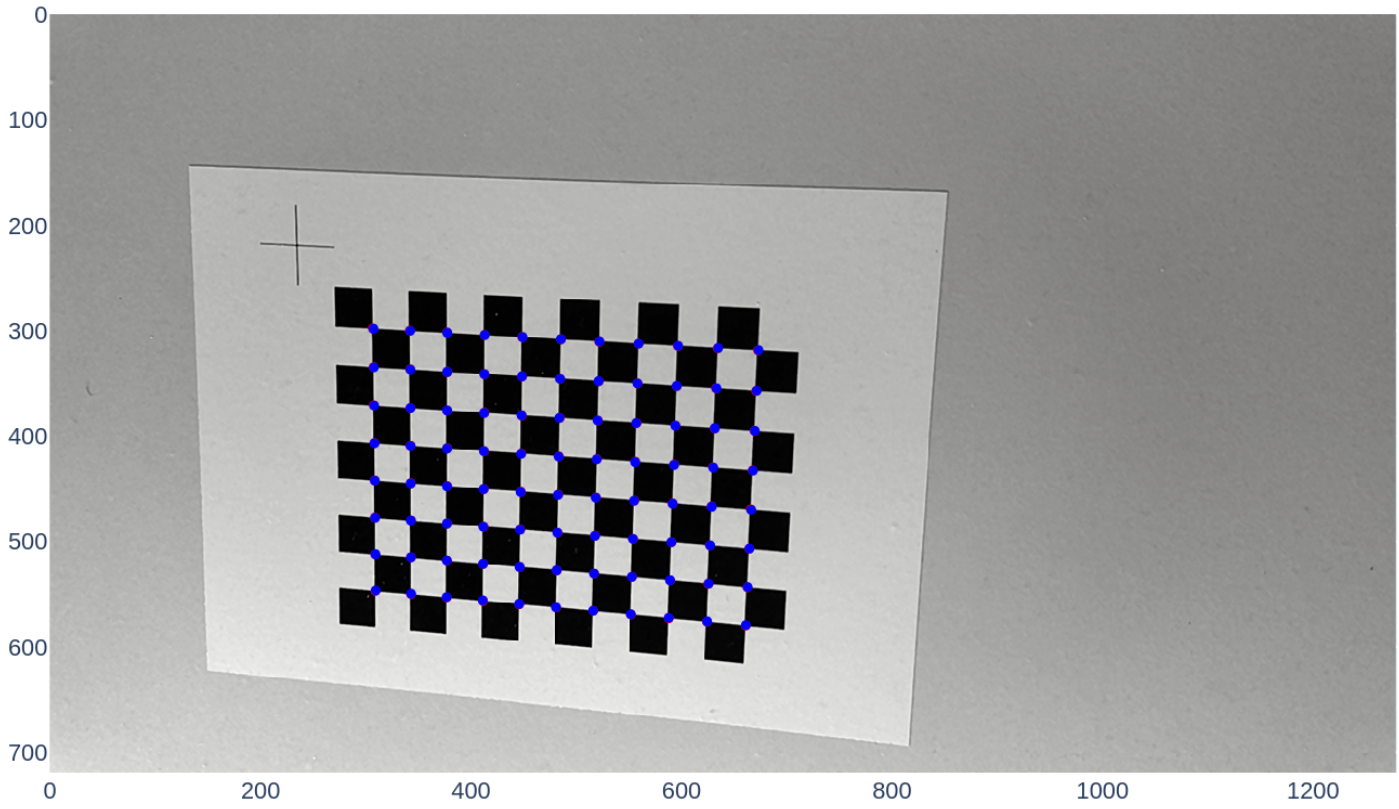


Figure 3: Maximum likelihood estimation: Corners (red) and projected points (blue)

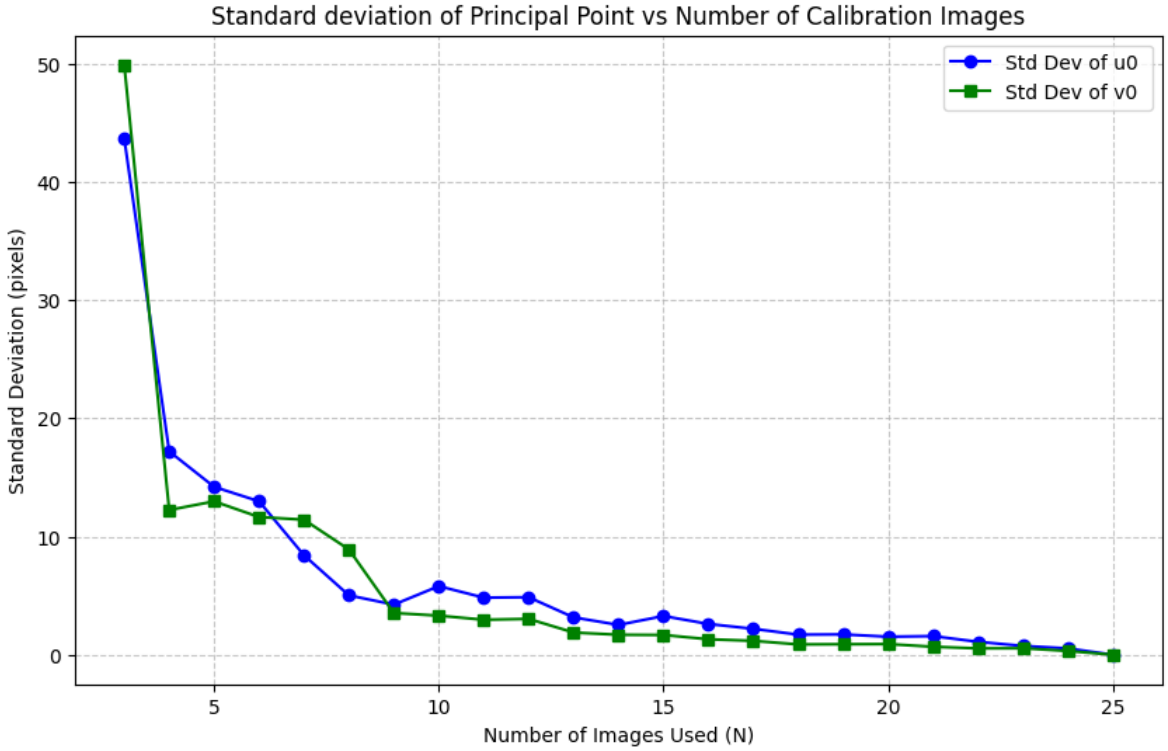


Figure 4: Standard deviation of the principal point

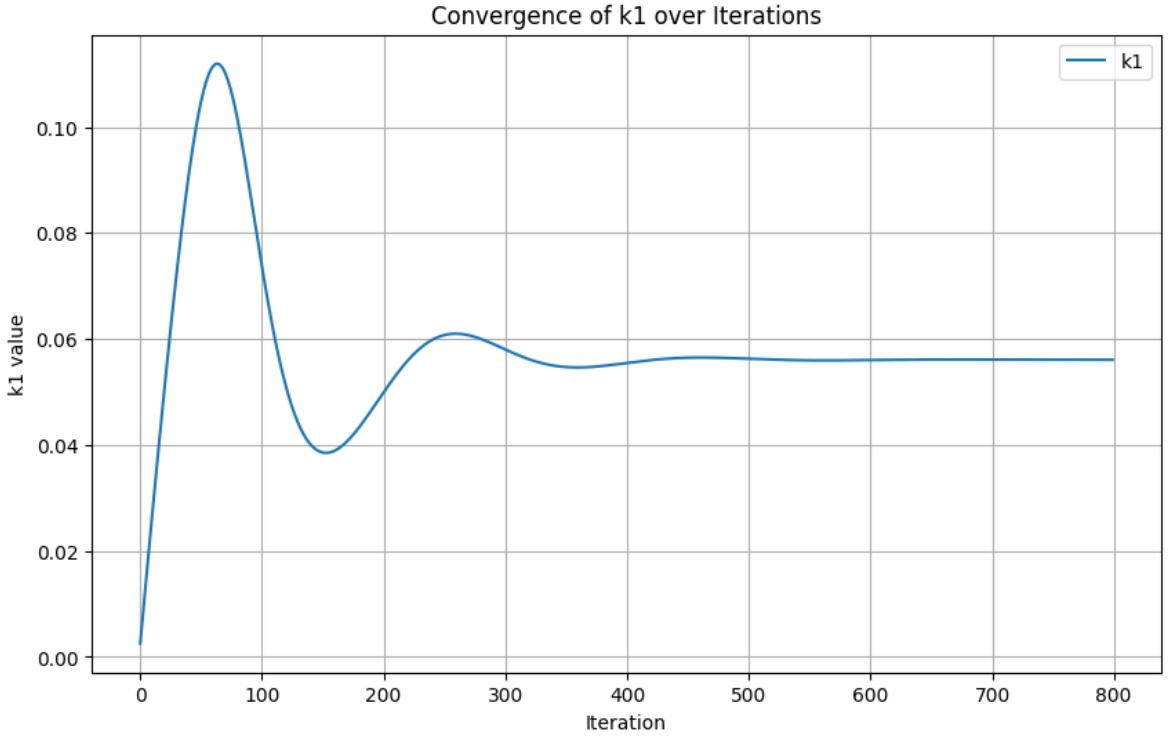


Figure 5: K1 estimation over iterations

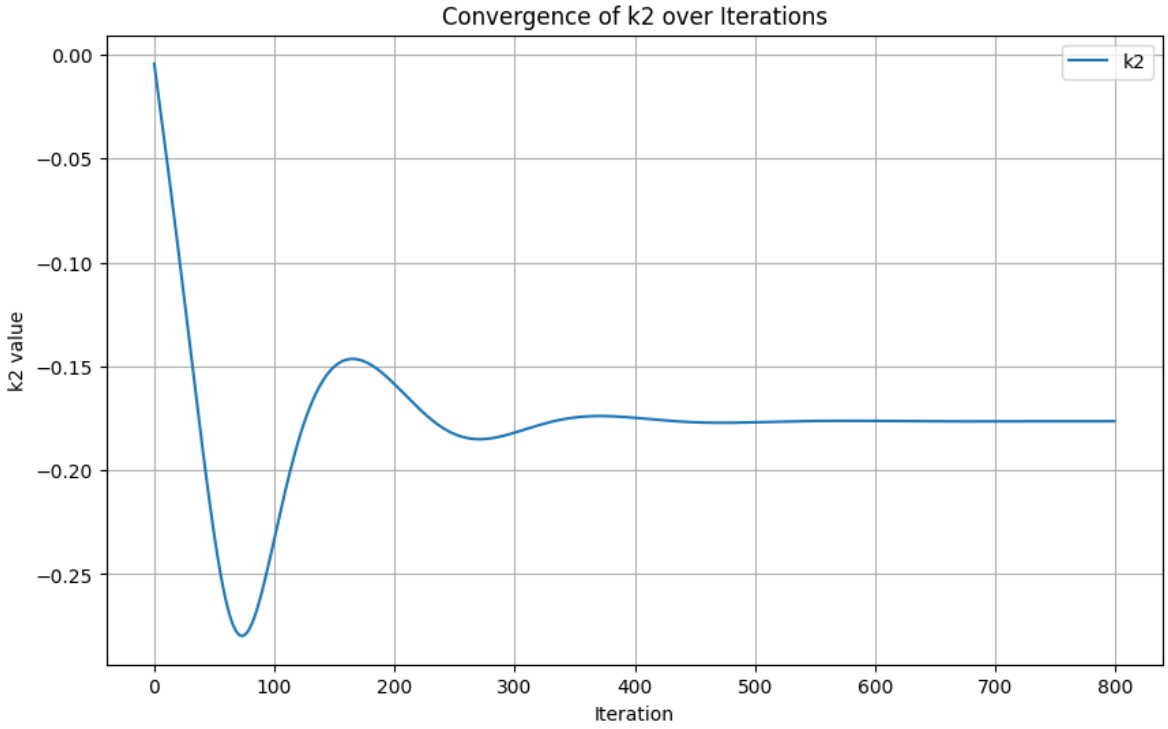


Figure 6: K2 estimation over iterations

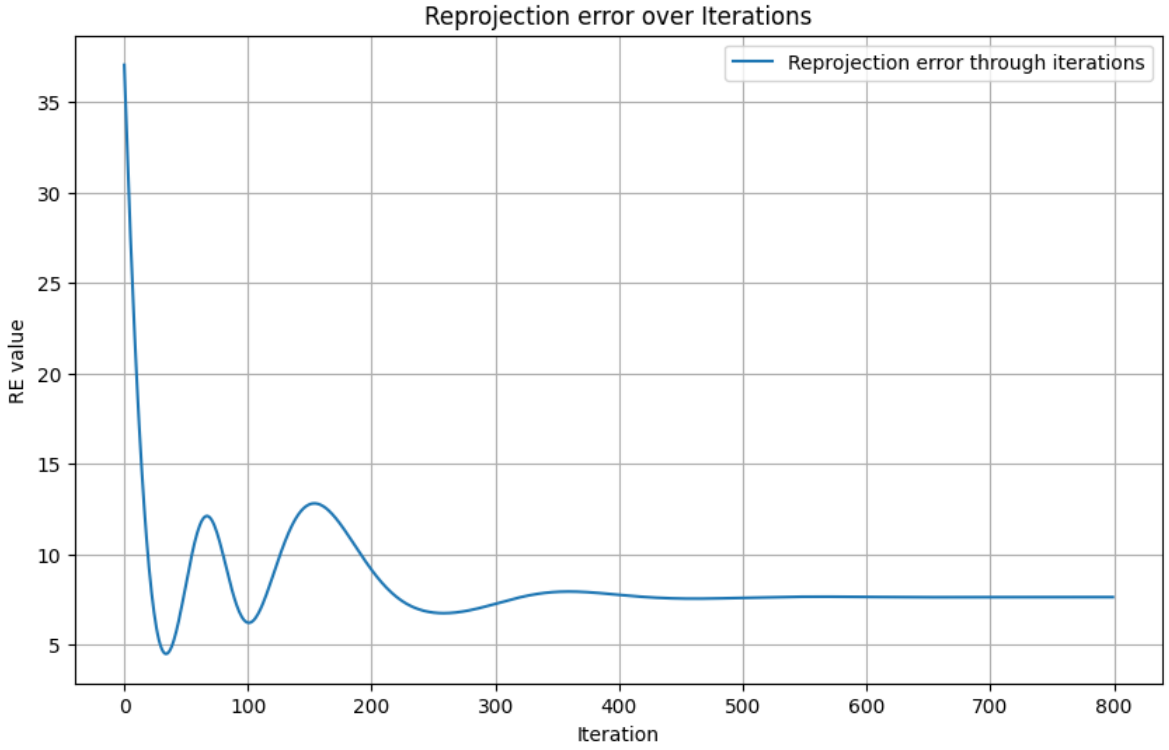


Figure 7: The Reprojection error over iterations

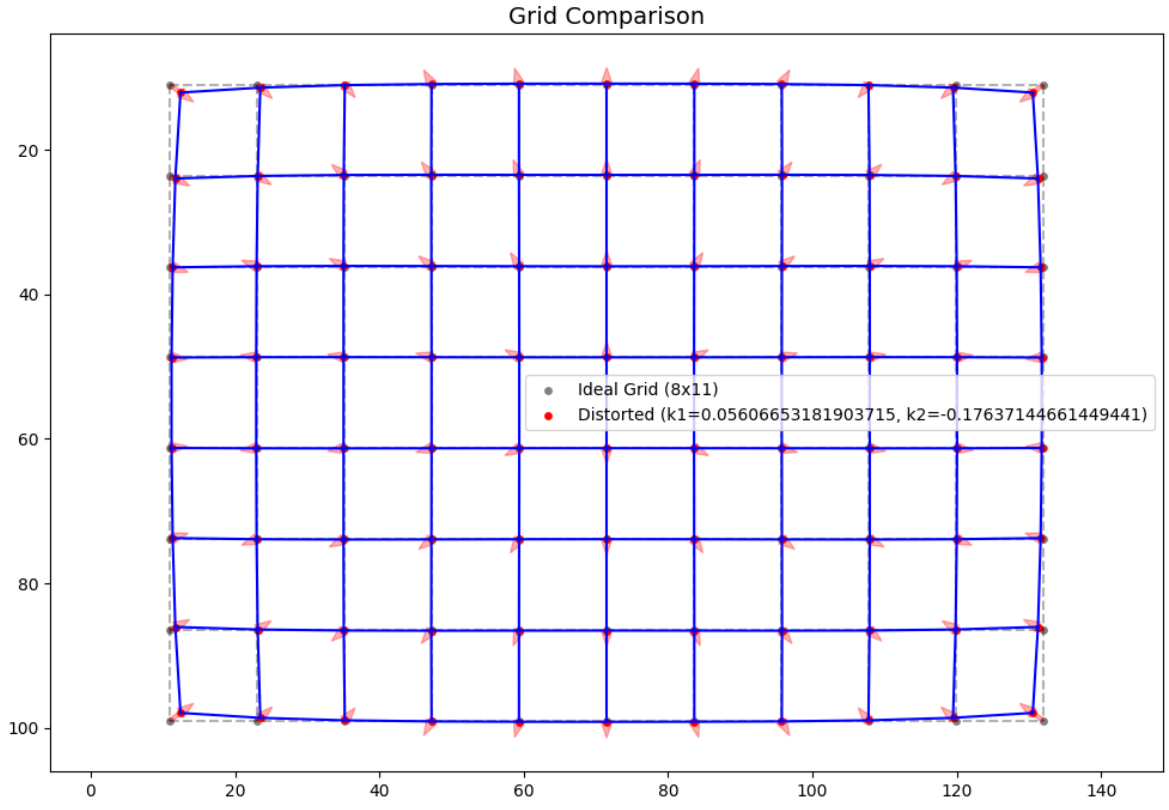


Figure 8: Undistorted grid VS distorted

References

[1] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.

[2] F. A. Pellegrino, "Slides cvpr2024 03 camera calibration." Slides for the Course Computer Vision and Pattern Recognition, 2024/2025.